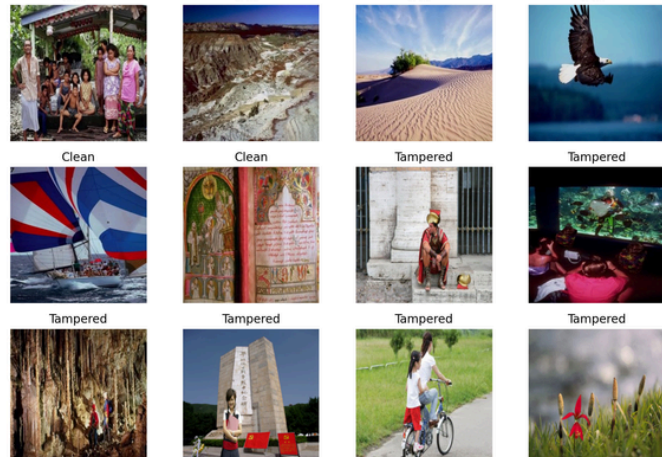


AIShield - A Preprocessing Defence Layer for Detecting Manipulated Inputs



AIShield enhances AI security by identifying adversarially manipulated images using a fine-tuned EfficientNet model with robust detection and interpretability methods.

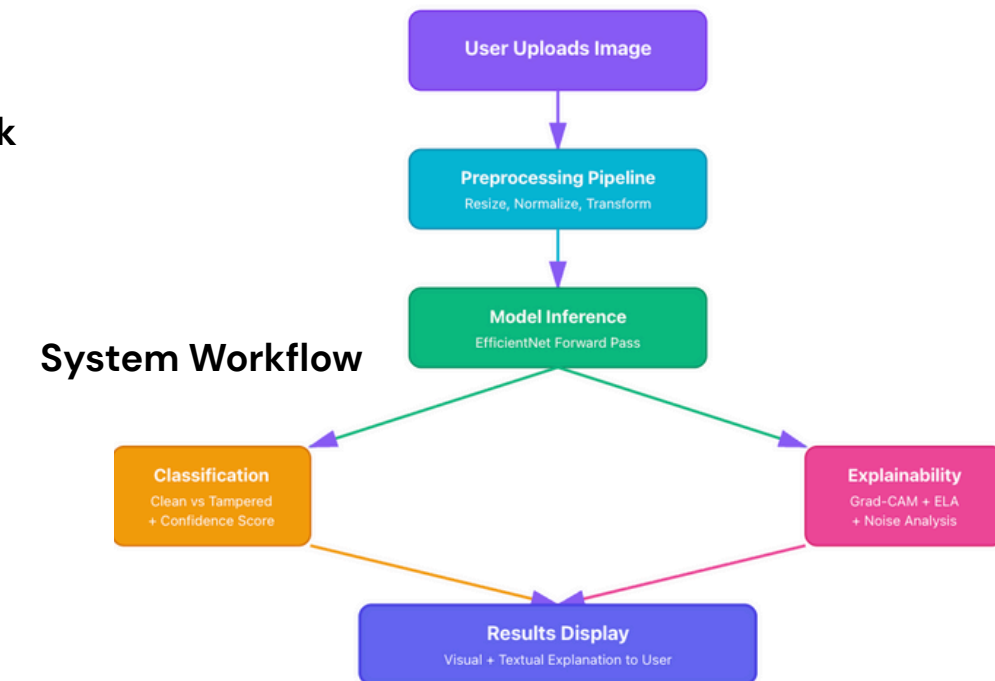
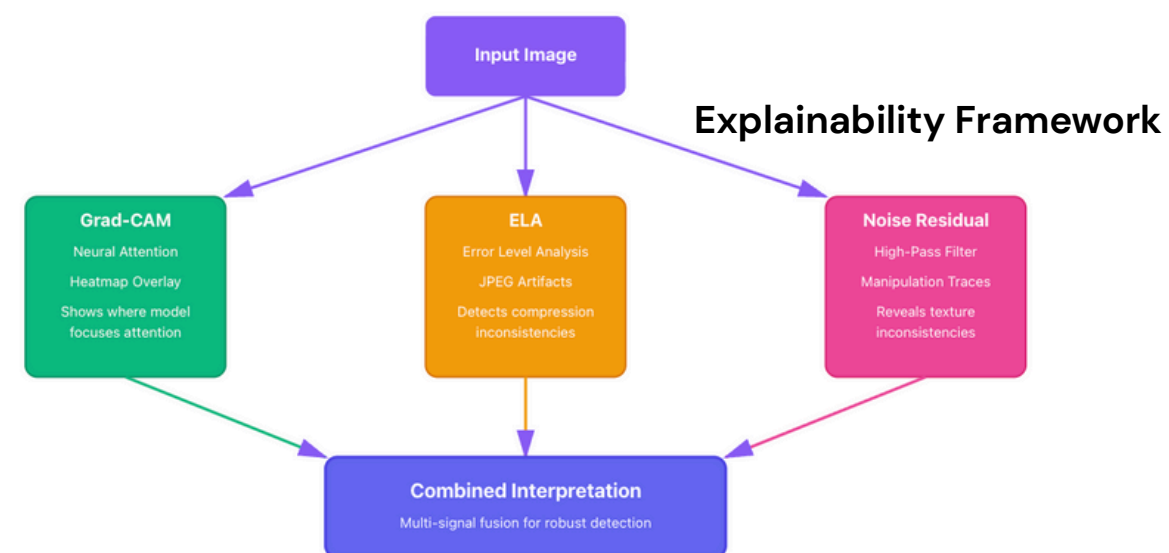
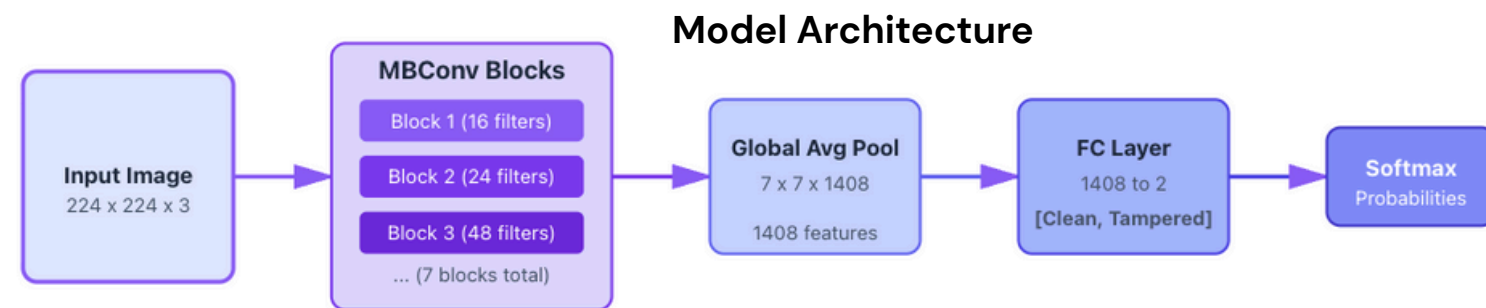
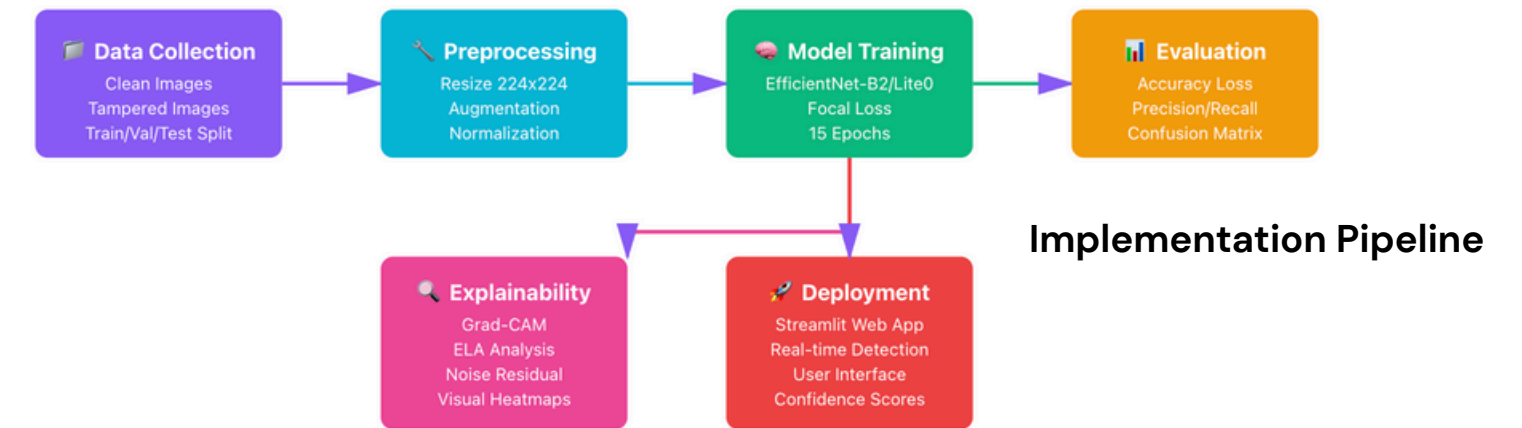
Dataset: 7,497 Clean Images | 5,117 Tampered Images.



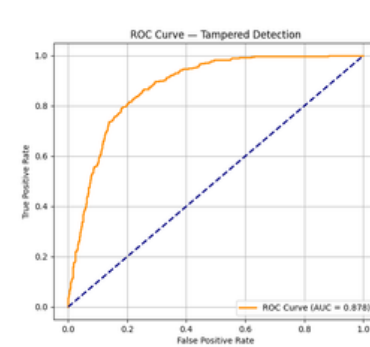
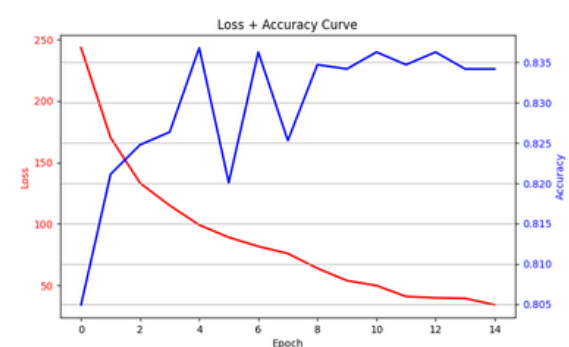
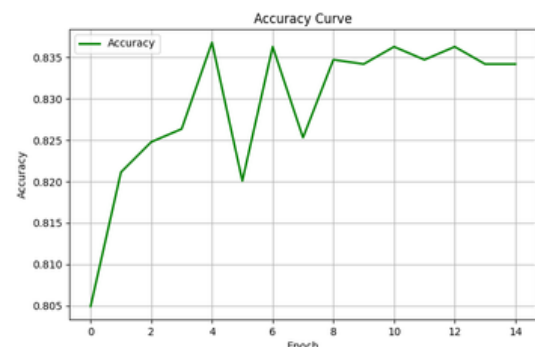
CASIA Image Tampering Dataset
A benchmark collection of real vs. manipulated images for digital forensics research.

Artificial intelligence models are often vulnerable to adversarial or tampered inputs; small, subtle modifications that can mislead the model into making incorrect predictions.

AIShield introduces a preprocessing defense layer that automatically detects and filters manipulated images before they are passed to the primary AI model.



Result



Metric	Class 0	Class 1
Precision	0.93	0.57
Recall	0.84	0.78
F1-Score	0.88	0.66
Support	1499	413

Metric	Score
Accuracy	0.82
Macro Avg F1	0.77
Weighted Avg	0.83